Project Report: Residual LSTM Correction Model for Hydrological Discharge Forecasting

Project Title: Hybrid Residual LSTM for Improving TOPKAPI Discharge Predictions (Casalecchio Basin)

Date: May 2026

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1. Executive Summary

This project developed a **hybrid Residual Correction LSTM** model that uses the physics-based TOPKAPI model as a baseline and trains an LSTM to learn and correct the residual error (Observed QM - TOPKAPI Q).

Key Result:

- **Residual LSTM** achieved **MAE = 8.63 m^3/s** and **RMSE = 16.97 m^3/s** on the test set.
- This represents a **~26% improvement** over the standalone TOPKAPI model.

This approach combines the strengths of a physically-based model with the error-learning capability of deep learning.

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2. Dataset Information

- **Basin**: Casalecchio
- **Time Period**: January 2013 – January 2026
- **Temporal Resolution**: Hourly
- **Total Records**: 113,953 hours (~13 years)
- **Target Variable**: Residual = QM - Q (Observed minus TOPKAPI)

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3. Features Used (14 Features)

- TOPKAPI predicted discharge (Q)
- Meteorological: Prec, Rain, Temp
- Evaporation & Snow: Evap, Snow, YSnow, EnSnow, SWE
- Soil & Surface: Soil, SoilSat, Perco, Surf, Inf2Surf

Input Window: Past **48 hours**

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4. Model Architecture (Residual LSTM)

- **Type**: LSTM-based residual corrector
- **Architecture**:
  - 2-layer LSTM (hidden size = 64, dropout = 0.25)
  - Fully connected layers (64 → 32 → 1)
  - L2 regularization (weight_decay=1e-5)
- **Training**:
  - Loss: MSE on residual
  - Optimizer: Adam
  - Early stopping (patience = 10)

Final Prediction Formula:

Final Discharge = TOPKAPI_Q + LSTM_Predicted_Residual

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5. Performance Comparison

Model	MAE (m^3/s)	RMSE (m^3/s)	Improvement vs TOPKAPI
Residual LSTM	**8.63**	**16.97**	**+25.7%**
Pure TCN	8.33	17.23	+28.3%
TOPKAPI (Baseline)	11.62	19.99	-
Direct LSTM (Prec+Temp only)	16.46	37.16	-

Conclusion: The Residual LSTM is a strong hybrid model that effectively corrects systematic biases in the TOPKAPI model.

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6. Project Structure

text

```
hydrology_residual_lstm/
├── data/
│   └── 425.sbs.ts
├── results/              # Predictions, plots, full comparison
├── models/               # best_residual_lstm.pth, scalers
├── scripts/
│   ├── data_preparation_residual.py
│   ├── train_residual_lstm.py
│   └── evaluate_residual.py
├── README.md
└── requirements.txt
```

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7. How to Reproduce

1. Place 425.sbs.ts in the data/ folder
2. Run data preparation:

Bash

```
python scripts\data_preparation_residual.py
```

3. Train the model:

Bash

```
python scripts\train_residual_lstm.py
```

4. Evaluate:

Bash

```
python scripts\evaluate_residual.py
```

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8. Advantages of Residual Approach

- Leverages physical knowledge from TOPKAPI
- Easier to train (learns only the error)

- More stable and interpretable than pure black-box models
- Good performance even with moderate data size